**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 288 · Symmetric and asymmetric encryption

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Which technique protects data against casual interception?

Q2. What does a network proxy do?

Q3. What is the difference between confidentiality and integrity?

Q4. Rising-edge D flip-flop Q=0, D=1 just before next edge. What happens to Q, and when?

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

symmetric — Shared-secret encryption.

asymmetric — Public/private-key cryptography.

public key — Distributable key in a pair.

private key — Secret member of key pair.

TLS — Protocol establishing secure authenticated channels.

## New learning

Understand — Symmetric encryption uses the same secret key for encryption and decryption. The difficulty is sharing that key securely.

Trace — Asymmetric encryption uses a public/private key pair. Public-key encryption for confidentiality uses a recipient's public key to encrypt and their private key to decrypt.

Evaluate — Real secure connections typically use a hybrid protocol: authenticated asymmetric cryptography to establish secrets, then efficient symmetric encryption of application data.

## Worked example

Alice and Bob already share the secret key K. Which key does each use to protect and recover a file?

1. Predict first — Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

2. Show method — Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

# We Do · Practise with guidance

Bob posts his public encryption key. Alice wants only Bob to read a message. Which key is used?

```
Encrypt(BobPublic, message)
Decrypt(BobPrivate, ciphertext)
```

Predict first

Alice encrypts to Bob's public key without needing to learn Bob's private key.

Show method

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.



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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Bob retains the private key and decrypts. The public key must be authenticated to prevent substitution.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Describe symmetric and asymmetric key arrangements. [6]

### Question 2

Explain how a secure website can combine asymmetric and symmetric methods. [6]

### Question 3

Evaluate symmetric-only encryption for a school communicating with 200 external partners. [6]

## Exit Ticket · Answer before checking

A public key must be kept secret to use public-key encryption.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Encryption.

Q2. Acts as an intermediary for requests/responses.

Q3. Confidentiality prevents unauthorised reading; integrity detects unauthorised changes.

Q4. Q changes to 1 on the next rising edge, not simply when D changes.

### You Do pages · Independent questions

Q1. Symmetric encrypt/decrypt use shared secret [2]; asymmetric uses public/private pair [2], encrypt to recipient's public key [1], private key decrypts [1].

Q2. Server identity authenticated via certificate [2]; authenticated key agreement establishes session secret [2]; symmetric session encryption protects bulk traffic efficiently [2].

Q3. Fast for bulk data [1]; each shared secret needs secure exchange [2]; managing many unique shared keys is difficult [1]; asymmetric authenticated exchange/hybrid design improves distribution [2].

### Exit Ticket · Model answer

False. The paired private key is secret; the public key needs authentic distribution.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**